

4. Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions.

CODE:

```
abhinav@Abhinav: × abhinav@Abhinav:
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        sleep(10);
    }
    else
    {
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());

        wait(NULL);
    }

    return 0;
}
```

Output:

```
abhinav@Abhinav:~/mnt/c/Users/HP/Downloads/OSSP-Project$ gcc pid.c -o program
abhinav@Abhinav:~/mnt/c/Users/HP/Downloads/OSSP-Project$ ./program
-----
Child Process
PID : 1091
PID : 1092
PPID : 1091
```